

Business Requirements Document - Hospital Management System

Version 1.0 | Derived from HMS Product Requirements Document v1.0

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DOCUMENT PURPOSE

This BRD translates the HMS product definition into business needs, stakeholders, processes, business rules, functional business requirements, scope boundaries and measurable outcomes. It is the bridge between the product vision and detailed functional/technical specifications.

Relationship to PRD: The PRD defines what HMS is as a product. This BRD explains why the business needs it, who uses it, how the major business processes operate, and what the system must enable at business level.

1. Document Control

Item	Details
Document	Business Requirements Document Hospital Management System
Version	1.0
Source	HMS Product Requirements Document v1.0
Product	Hospital Management System (HMS)
Primary audience	Hospital stakeholders, management, product, UI/UX, engineering, QA and implementation teams
Document purpose	Define business objectives, stakeholders, workflows, requirements, rules, scope and acceptance expectations.

Document governance

- The BRD is the baseline for business scope and workflow expectations.
- Detailed UI behaviour, API contracts, database schemas and test cases should be derived from the BRD and PRD.
- Any new requirement outside the approved scope should be handled as a change request.

SOURCE ALIGNMENT: The product is intended as a connected hospital ecosystem rather than a collection of isolated departmental tools.

2. Business Overview

HMS is a centralized hospital management platform intended to connect the hospital's clinical, administrative, financial, diagnostic and patient-facing operations. It gives each authorized role a dedicated workspace while keeping shared hospital information connected.

The business value comes from reducing fragmented workflows. A patient registered by reception should become visible to the appropriate clinical workflow; a doctor's prescription should support pharmacy/laboratory actions; diagnostic results should return to the patient's record; billing should reflect services; and patients should be able to access approved information digitally.

BUSINESS OUTCOME: HMS should create a single operational picture of the hospital while preserving role-based control over sensitive information.

3. Business Problem

Business issue	Impact
Fragmented information	Patient and operational information can be spread across departments, creating duplication and delays.
Manual coordination	Reception, doctors, nurses, diagnostics and billing depend on handoffs that can be slow or error-prone.
Limited visibility	Management may lack a real-time view of appointments, occupancy, staff availability, critical alerts and revenue.
Patient friction	Patients may need to repeatedly provide information or depend on staff for appointments, reports and billing information.

Business issue	Impact
Data integrity risk	Duplicate records and inconsistent updates can reduce confidence in operational and clinical information.
Limited traceability	Without centralized logs, it is difficult to understand who changed or accessed important information.

4. Business Objectives

- Create one connected digital platform for core hospital operations.
- Improve the patient journey from registration through consultation, diagnostics, billing, discharge and follow-up.
- Reduce duplicate data entry and manual coordination between departments.
- Improve visibility of beds, appointments, staff, patients, diagnostics and financial activity.
- Provide clinicians with relevant patient information at the point of care.
- Enable patients to access their own approved healthcare information and services.
- Improve accountability through role-based permissions and audit history.
- Create a scalable foundation for future integrations and enterprise expansion.

5. Stakeholders

Stakeholder	Business interest
Hospital Ownership / Management	Business oversight, performance, revenue, compliance and strategic decisions.
Super Admin / Platform Owner	Platform governance, hospital/tenant administration, users, security and system-wide oversight.
Hospital Admin	Daily hospital operations, departments, staff, beds, patients, finance and reporting.
Doctors	Clinical consultation, diagnosis, prescription, investigations and follow-up.
Nurses	Patient monitoring, medication administration, nursing notes and ward care.

Stakeholder	Business interest
Receptionists	Registration, appointments, check-in, OPD and admissions.
Billing Staff	Invoices, payments, insurance/TPA, refunds and financial reporting.
Laboratory Staff	Test orders, samples, processing, results and reports.
Patients / Guardians	Appointments, records, reports, prescriptions, payments and communication.

6. User Roles & Business Responsibility

Role	Primary business responsibility
Super Admin	Govern the platform and participating hospitals/tenants; manage global access, security, subscriptions and system-level reporting.
Hospital Admin	Operate and administer an individual hospital across departments, people, patients, resources and finances.
Doctor	Deliver clinical care and maintain authorized patient clinical records.
Nurse	Execute nursing and ward-care activities for assigned patients.
Receptionist	Manage patient-facing front-desk and admission workflows.
Billing Staff	Manage hospital service billing, collections, claims and refunds.
Laboratory Staff	Execute diagnostic testing workflows and publish results.
Patient	Use self-service healthcare functions and access the patient's own approved information.

7. Business Process Architecture

- **Front Desk:** Patient identity → registration → appointment/walk-in → check-in → queue.
- **Clinical:** Queue → consultation → clinical record → diagnosis → prescription/investigation → follow-up.
- **Nursing:** Assigned patient → vitals/notes → medication administration → escalation → handover.
- **Diagnostics:** Order → sample/request → processing → result → report → clinician/patient access.
- **Inpatient:** Admission → ward/bed allocation → treatment → transfer/monitoring → discharge.
- **Financial:** Service → bill → payment/insurance → receipt/refund → financial reporting.
- **Patient:** Account → appointment → care → reports/prescriptions → billing → follow-up.

8. End-to-End Patient Journey

Step	Stage	Business requirement
01	Registration	Create or identify the patient record and maintain a unique hospital identity.
02	Appointment / Walk-in	Schedule a consultation or process the patient as a walk-in.
03	Check-in	Record arrival and place the patient into the appropriate queue.
04	Consultation	Doctor reviews relevant history and records consultation, diagnosis and treatment.
05	Diagnostics	Create and process laboratory/radiology investigations and publish results.
06	Treatment	Prescription, nursing care

Step	Stage	Business requirement
		and medication workflows support treatment.
07	Admission	If required, allocate ward/bed and manage inpatient treatment.
08	Billing	Generate applicable charges, collect payment and process insurance/TPA.
09	Discharge	Complete discharge documentation and financial closure where applicable.
10	Follow-up	Schedule follow-up care and continue patient engagement through the portal.

9. Business Requirements

ID	Business area	Requirement
BR-01	Hospital Administration	The system must allow authorized administrators to configure and manage hospital departments, staff, operational resources, settings and hospital-level controls.
BR-02	Patient Management	The system must maintain a central patient identity and allow authorized users to access the information required for their responsibilities.
BR-03	Appointments	The system must support doctor schedules, appointment booking, rescheduling, cancellation, check-in and queue

ID	Business area	Requirement
		management.
BR-04	OPD	The system must support outpatient registration, consultation flow and daily OPD operations.
BR-05	IPD	The system must support inpatient admission, ward/bed allocation, transfers, treatment lifecycle and discharge.
BR-06	Clinical Care	The system must support consultation, diagnosis, vitals, prescriptions, nursing notes and clinical follow-up.
BR-07	Diagnostics	The system must support laboratory and radiology orders, sample/testing workflows, result management and report availability.
BR-08	Pharmacy	The system must support prescription fulfillment, medicine inventory, dispensing, stock monitoring and pharmacy billing where enabled.
BR-09	Billing	The system must support service billing, invoices, payments, receipts, discounts, refunds and financial reporting.
BR-10	Insurance/TPA	The system must support insurance verification, pre-authorization, claim tracking and settlement workflows where applicable.
BR-11	Hospital Resources	The system must support

ID	Business area	Requirement
		inventory, assets, ambulances, shifts, attendance and visitor management.
BR-12	Communication	The system must support internal communication, notifications, alerts, reminders and patient-facing updates.
BR-13	Reporting	The system must provide operational, clinical, financial and administrative reporting to authorized users.
BR-14	Security & Audit	The system must enforce role-based access and retain traceability of important system actions.
BR-15	Patient Portal	The system must provide patients with secure access to appointments, records, prescriptions, reports, bills and support functions.

10. Business Rules

Rule	Area	Business rule
BRL-01	Patient identity	A patient should have one primary hospital identity/UHID within the applicable hospital environment.
BRL-02	Access control	Users may access only modules, records and actions permitted by their role and organizational scope.
BRL-03	Clinical responsibility	Clinical records and

Rule	Area	Business rule
		treatment decisions remain the responsibility of authorized healthcare professionals.
BRL-04	Auditability	Important record changes, access events, permissions and financial actions must be traceable.
BRL-05	Queue integrity	Check-in and queue status must be reflected consistently for the relevant receptionist and clinical users.
BRL-06	Order traceability	Diagnostic orders must remain associated with the originating patient and authorized ordering workflow.
BRL-07	Billing traceability	Invoices, payments, refunds and adjustments must retain transaction history.
BRL-08	Sensitive information	Patient information must not be exposed to roles that do not require it for their business function.
BRL-09	Notification relevance	Notifications should be routed to responsible users rather than broadcast indiscriminately.
BRL-10	Change control	New functionality outside approved scope requires explicit business approval.

11. Role-Based Business Workflows

Role	Primary workflow
Receptionist	Register patient → book appointment → check-in → queue/token → coordinate admission if required.
Doctor	Open waiting patient → review record → consult → diagnose → prescribe/order tests → schedule follow-up.
Nurse	View assigned patients → record vitals → administer medication → add notes → escalate critical condition → handover.
Laboratory	Receive order → collect/identify sample → process test → verify result → publish report → notify relevant users.
Billing	Review services → generate invoice → collect payment/claim → issue receipt → process refund/settlement if applicable.
Hospital Admin	Monitor dashboard → manage departments/staff → oversee beds/appointments → review finance/reports → manage operational alerts.
Super Admin	Monitor hospitals/tenants/subscriptions → manage access/security → review platform analytics → govern audit system activity.
Patient	Maintain profile → book appointment → attend care → view prescription/report → pay bill → manage follow-up.

12. Cross-Department Integration Requirements

Integration	Expected business behaviour
Reception → Doctor	Check-in and queue status should become available to the clinical workflow.
Doctor → Patient Record	Consultation, diagnosis, prescriptions and orders should update the patient's connected record.

Integration	Expected business behaviour
Doctor → Laboratory	Investigation orders should become available to laboratory staff for processing.
Laboratory → Doctor/Patient	Published reports should become accessible to authorized recipients.
Doctor → Pharmacy	Digital prescriptions should support pharmacy fulfillment where pharmacy module is enabled.
Clinical → Billing	Chargeable services should be available to billing workflows where configured.
Billing → Patient	Invoices, payments and receipts should be visible to the patient where permitted.
System → Users	Relevant registration, report, appointment, billing and critical alerts should generate appropriate notifications.

SINGLE SOURCE OF TRUTH: The system should behave as one connected platform: information created in one workflow should be reusable by authorized downstream workflows rather than being manually recreated.

13. Business Reporting Requirements

- Hospital operational dashboard and KPI summaries.
- Appointment, OPD and queue reports.
- Patient registration and admission reports.
- Bed occupancy and ward reports.
- Doctor/staff availability, duty and attendance reports.
- Clinical activity and patient monitoring reports.
- Laboratory and diagnostic workload/results reports.
- Pharmacy and inventory reports.
- Billing, collection, outstanding, refund and insurance/TPA reports.
- Audit and user activity reports.
- Patient-facing summaries and downloadable records where authorized.

14. Notifications & Alerts

Event	Business action	Typical recipients
Appointment	Confirm/remind/reschedule/cancel appointment.	Patient, Receptionist, Doctor
Patient registration	Provide account/activation or registration information.	Patient, Receptionist
Critical patient	Escalate a critical or emergency condition.	Doctor, Nurse, relevant clinical staff
Diagnostic report	Notify when a report is available; critical results may require escalation.	Doctor, Patient where permitted
Medication	Notify relevant nursing staff about due/missed medication events.	Nurse
Billing	Notify about invoice, payment, outstanding amount or claim status.	Patient, Billing
Operational alert	Notify responsible hospital team about bed, department or emergency issues.	Hospital Admin / relevant staff

15. Security, Privacy & Compliance Business Requirements

- Access to patient and hospital data must be controlled by role, hospital/tenant scope and permissions.
- Sensitive actions must be traceable through audit logs.
- Authentication and session security must protect user accounts.
- Patient data must be handled according to applicable privacy and healthcare regulations for the deployment geography.
- The product must not claim regulatory certification solely because a feature exists; deployment-specific compliance validation remains required.
- Backups, recovery, monitoring and operational security controls should be part of the implementation plan.

COMPLIANCE BOUNDARY: The BRD defines security and privacy expectations; it does not by itself certify HMS as HIPAA/NABH or any other regulatory-compliant system.

16. Scope Definition

In Scope

- Hospital administration and role-based access.
- Patient, appointment, OPD and IPD management.
- Wards, beds, emergency and core hospital operations.
- Doctor and nurse clinical workflows.
- Laboratory and radiology coordination.
- Pharmacy, inventory, asset and resource workflows.
- Billing, payments, insurance/TPA, refunds and reporting.
- Patient portal and patient self-service.
- Notifications, communication, audit logs, reports and global search.

Out of Scope unless separately approved

- Autonomous medical diagnosis or treatment.
- Replacing licensed medical professionals.
- Unapproved external integrations.
- Regulatory certification itself.
- Features not mapped to an approved product/module requirement.
- Custom workflows requested by a specific hospital that change the baseline product without change approval.

17. Assumptions & Dependencies

Dependency	Business assumption
Hospital data	Accurate patient, doctor, staff, department and service data will be available for onboarding/configuration.
User roles	Hospital management will define which staff members receive which roles and permissions.
Clinical workflows	Final clinical workflows will be validated with

Dependency	Business assumption
	actual hospital stakeholders before production.
Integrations	External systems such as payment, messaging, diagnostic or insurance systems depend on available APIs and credentials.
Regulatory	Deployment-specific legal, privacy and healthcare requirements will be validated for the target geography.
Infrastructure	Production hosting, backup, monitoring, email/SMS and file storage services will be provisioned separately.
Change requests	Requirements introduced after baseline approval may affect schedule and cost.

18. Business Acceptance Criteria

- A patient can be registered and identified consistently across the core hospital workflows.
- A receptionist can complete an appointment/check-in workflow and the clinical team can see the resulting patient queue.
- A doctor can conduct a consultation and maintain the patient's relevant clinical record.
- Diagnostic orders can move through the laboratory workflow and return as an accessible report.
- Prescription and nursing workflows can be completed without requiring duplicate patient entry.
- IPD patients can be associated with wards/beds and their status can be tracked.
- Billing can produce an invoice, record payment and provide a transaction/receipt history.
- Patients can access only their own authorized portal information.
- Hospital administrators can access meaningful operational reports and dashboards.
- Unauthorized users cannot access restricted modules or patient information.
- Important operational and administrative actions are auditable.

19. Business Success Measures

Area	Target outcome
Efficiency	Less manual coordination and faster completion of common hospital workflows.
Patient experience	Lower waiting friction and greater self-service access.
Clinical continuity	Faster access to relevant history, orders and reports.
Financial control	Improved visibility of collections, outstanding balances and claims.
Operational visibility	Management can monitor hospital KPIs from centralized dashboards.
Data quality	Fewer duplicate records and fewer inconsistent operational updates.
Accountability	Important actions and changes are traceable to authorized users.

20. Business Release Priorities

Priority	Business release focus
P0 - Foundation	Authentication, roles/permissions, hospital setup, patient identity, core records and audit foundation.
P0 - Core Journey	Registration, appointments, check-in, OPD, consultation and basic EMR.
P0 - Clinical & Finance	Prescriptions, diagnostics, nursing basics, billing, payments and receipts.
P1 - Operations	IPD, wards, beds, pharmacy, inventory, emergency and notifications.
P1 - Patient Experience	Patient portal, reports, prescriptions, billing access and reminders.
P2 - Enterprise	Advanced analytics, OT, ambulance, assets, compliance and deeper integrations.

21. Change Control

The BRD establishes the baseline business scope. Changes should be evaluated for business value, affected roles, workflow impact, data impact, security/privacy impact, dependencies, effort and release priority before approval.

- New module or major workflow → formal change request.
- Change to role permissions → security/access review.
- Change to patient/clinical data → clinical and data-impact review.
- Change to billing/insurance → financial and audit review.
- External integration → dependency and security review.
- Regulatory requirement → compliance/legal validation.

22. Final Business Definition

THE BUSINESS CASE

HMS provides a unified operating environment for hospitals by connecting patient identity, front-desk operations, clinical care, nursing, diagnostics, pharmacy, inpatient operations, billing, administration and patient self-service into one controlled workflow.

The central business principle is simple: the patient's journey should remain continuous even when responsibility moves from one department to another. The system therefore exists to connect those handoffs, reduce repeated work, improve visibility and preserve trustworthy records.

BRD Status: Business Requirements Baseline v1.0